Lecture 9: Nested Parallelism

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Nested Parallelism

- Nested parallelism enables the programmer to create parallelism within a parallel execution
  - For example, it can combine task parallelism and data parallelism
  - Not every multithreading library allows this
- OpenMP has support for nested parallelism:
  - Create a parallel region inside of a parallel region
  - Or create a data parallel region (with a for--loop) inside of a task.

Nested Parallel Regions

```
int main() {

    omp_set_nested(1);

    #pragma omp parallel num_threads(2) {

        #pragma omp parallel num_threads(2)
        {
            #pragma omp parallel num_threads(2)
            {


            }
        }
    }
    return(0);
}
```

Enables nesting
export OMP_NESTED=TRUE

How many threads are there?

Combining Task and Data Parallelism

```
omp_set_nested(1);

#pragma omp parallel
  {

  }
  return 0;
}
```

In a parallel region, only one thread enters into single creates two tasks, one of them creates another parallel region with 4 threads.

2nd Assignment

- Game of Life
  - Cellular Automata
  - Very simple simulation
- Parallel Implementation of Game of Life
  - Start with an infinite 2D array of cells
  - Two possible states: live or dead
  - Each cell interacts with its 8 neighbors

Update Rules

Iterate over timesteps

```
for t in 0:T-1
     forall (i,j) in 1:N x 1:N
         nNeigh = number of live neighbors of World(t)[i,j]
         World(t+1)[i,j] = DEAD
         if World(t)[i,j] AND (nNeigh == 2 or nNeigh ==3) then
           World(t+1)[i,j] = LIVE
         else
           World(t+1)[i,j] = (nNeigh == 3)
         end if
     end forall
   end for
```

Compute live neighbors

Cannot do an in–place update, need two worlds: t and t+1

Algorithm

```
For each time step
    for I to N
        for J to N
            calculateNeighbors of CurrWorld(I,J)
            update (NextWorld(I,J))
    swapPointers
    plot CurrWorld using Gnuplot
```

Combining Data + Task Parallelism

- Plotter thread
  - Single thread plots the universe
- Update threads
  - Remaining threads update the universe